	Reasonable smooth curve with equal fringe separation
5a	Increasing current causes increasing mangetic flux (linkage) through the coil. (By Faraday's Law), increasing magnetic flux (linkage) induces e.m.f. in the coil. (By Lenz's Law), induced e.m.f. opposes the growth of current. Note: the induced e.m.f. will be opposite to the e.m.f. of the d.c. source. As a result, current increases at a slower rate.
5bi	It is the product of the magnetic flux density and the area normal to the field through which the field is passing.
5bii	Magnetic flux density by solenoid B = $\mu_o nI = 4\pi \times 10^{-7} \times (360 / 0.120) \times 2.5$ A flux $\phi = BA = B(\pi D^2/4) = B[\pi(0.088/\pi)^2/4]$ $= 5.808 \times 10^{-6}$ Wb $= 5.8 \times 10^{-6}$ Wb
5biii	In the time given, $\Delta\phi = 2 \times 5.8 \times 10^{-6}$ Wb Average e.m.f. = $N\Delta\phi/t = 96 \times 2 \times 5.8 \times 10^{-6} / 2.4 \times 10^{-3}$ $= 0.464$ V $= 0.46$ V
5biv	Alternating / sinusoidal e.m.f. With same frequency as supply
6a	for a wave, electron can 'collect' energy continuously hence, electron will always be emitted / electron will be emitted at all frequencies